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21CSC305P MACHINE LEARNING | PROJECT REVIEW 3

Presentation Script

Grouping Warehouse Products with K-Means Clustering

Team Cash guard (No. 17) | 13 October 2026 | 18 slides, 4 speakers

Who presents what

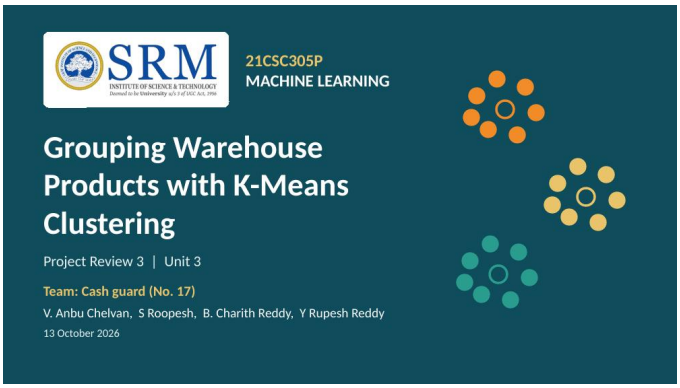
Speaker	Slides	What they explain	Time
1. S Roopesh	1 to 5	Title, the problem, key terms, Unit 3 solutions, our choice	about 5 min
2. B. Charith Reddy	6 to 10	Why not the others, how K-Means works, the real data, choosing k	about 5 min
3. Y Rupesh Reddy	11 to 15	Silhouette check, clusters to zones, the cluster graphs	about 5 min
4. V. Anbu Chelvan	16 to 18	How it solves the problem, the aisle and the walk, conclusion	about 3 min

HOW TO USE THIS SCRIPT

- One page for each slide. The table above shows which slides are yours.
- The **Say this** part is written the way you can speak it. Say it in your own words; do not read it word for word.
- Use the **Simple example** if the idea is not clear, or if you have time.
- The last line of your last slide hands over to the next speaker by name.
- The whole presentation takes about 15 minutes. Keep to under one minute per slide.
- The table, the curve and the four clusters come from real data (Olist's public orders). Only the walking distance is a simple model.

Speaker 1: S Roopesh Slide 1 of 18 | under 1 minute

Title slide



TERMS TO KNOW

K-Means clustering: A method that divides data into k groups of similar items.

Unit 3: The part of our syllabus on clustering, mixtures of Gaussians, factor analysis and PCA.

SAY THIS

Good morning everyone. We are team **Cash guard**. I am Roopesh, and my teammates are Charith Reddy, Rupesh Reddy and Anbu Chelvan.

This is our Project Review 3 for Machine Learning, 21CSC305P. Our topic is **grouping warehouse products with K-Means clustering**.

We studied the solutions given in Unit 3 and selected K-Means for our problem. I will explain the problem and the solution we chose. Charith will show how K-Means works on real data. Rupesh will show the clusters. Anbu will show how it solves the problem and conclude.

SIMPLE EXAMPLE

Think of a supermarket. Milk and bread are kept where everyone can reach them quickly. We want a warehouse to place its products in the same smart way.

IF THE REVIEWER ASKS

Q: What is your project in one line?

A: We group similar warehouse products with K-Means so that they can be placed smartly and workers walk less.

Our problem statement



TERMS TO KNOW

SKU: Stock Keeping Unit: one unique product type.

Partition: Divide into groups so that each item is in exactly one group.

Order fulfilment: Collecting the items of an order and sending it out.

Inventory placement: Deciding where each product is stored.

SAY THIS

A warehouse stores thousands of different products. Each product type is called an **SKU**. If products are kept in random places, workers walk a long way to collect the items of each order.

Our problem asks us to **partition**, that is divide, the SKUs into groups of similar products. We group them using four things. One, **pick frequency**: how often it is picked. Two, **seasonal demand**: how fast it sells in each season. Three, **physical size**: length, width, height and weight. Four, **handling needs**: fragile, heavy or needing special care.

Once we have the groups, the manager can decide where to keep each group. The goal is better inventory placement and less travel time.

SIMPLE EXAMPLE

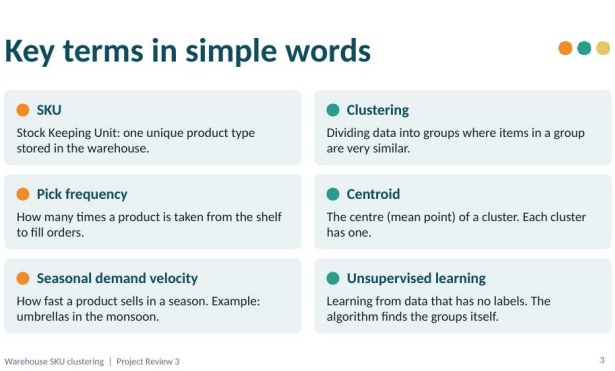
At home, the things we use every day are kept within reach, and the things we use once a year are kept on the top shelf. A warehouse has too many products to decide this by hand, so we let the data decide.

IF THE REVIEWER ASKS

Q: Why does placement matter?

A: Walking between shelves is a big part of a picker's work. If fast items are near the packing area, every order takes less walking.

Key terms in simple words



TERMS TO KNOW

Feature: A measurable property, such as size or pick frequency.

Label: The correct answer given along with the data. We have none.

Cluster: A group of similar data points.

Centroid: The mean point of a cluster.

SAY THIS

Before the solution, six simple terms. **SKU** is one unique product type. A 1 litre water bottle is one SKU and a 500 ml bottle is another SKU.

Pick frequency is how many times a product is taken from the shelf. **Seasonal demand velocity** is how fast it sells in a season: umbrellas in the monsoon, sweaters in winter.

Clustering means dividing data into groups where the items in a group are very similar. The **centroid** is the centre, or average point, of a cluster. **Unsupervised learning** means the data has no labels, so the algorithm finds the groups by itself.

Next, the solutions we reviewed.

SIMPLE EXAMPLE

Clustering is like sorting a basket of mixed fruits. Nobody gives us the names, but we still put apples with apples and oranges with oranges because they look alike.

IF THE REVIEWER ASKS

Q: Why is this unsupervised learning?

A: Nobody has told us which SKU belongs to which group. There are no labels, so the algorithm must find the groups.

Unit 3 solutions we reviewed



TERMS TO KNOW

Hard clustering: Each item belongs to exactly one group.

Soft clustering: Each item gets a probability for every group.

Dendrogram: A tree diagram that shows how clusters join.

EM algorithm: Expectation-Maximization: guess, improve, and repeat.

PCA: Principal Component Analysis: fewer features, same information as far as possible.

SAY THIS

Unit 3 gives us four kinds of solutions.

First, **K-Means clustering**. It splits the data into k groups, and each item joins the nearest centre. Second, **Mixtures of Gaussians** with the EM algorithm. This is soft clustering: each item gets a probability for every group.

Third, **hierarchical clustering**. It builds a tree of clusters, called a dendrogram, by merging or splitting step by step. Fourth, **PCA and factor analysis**. These reduce many features to a few, but they do not make groups.

We compared all four with our problem and selected K-Means.

SIMPLE EXAMPLE

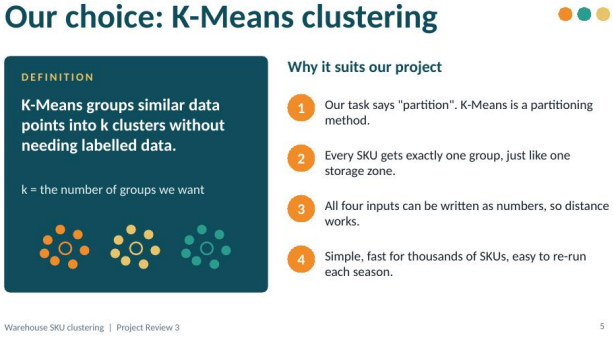
Hard clustering says: this student is in section A. Soft clustering says: 70% section A and 30% section B. A product on a shelf needs the first kind of answer.

IF THE REVIEWER ASKS

Q: Which of these make groups?

A: K-Means, mixtures of Gaussians and hierarchical clustering make groups. PCA and factor analysis only reduce the number of features.

Our choice: K-Means clustering



TERMS TO KNOW

k: The number of clusters we ask for.

Partitioning method: A method that divides data into separate groups.

Labelled data: Data that already has the correct answer attached.

Distance: A number that says how different two items are.

SAY THIS

Our selected solution is K-Means. **K-Means groups similar data points into k clusters without needing labelled data.** Here k is simply the number of groups we want.

Why does it suit our project? Reason one: our problem statement says partition, and K-Means is a partitioning method. Reason two: every SKU gets exactly one group, just as a product is kept in one storage zone.

Reason three: all four inputs can be written as numbers, so the algorithm can measure the distance between products. Reason four: it is simple and fast even for thousands of SKUs, and we can run it again every season.

Now Charith will explain why we did not pick the other methods.

SIMPLE EXAMPLE

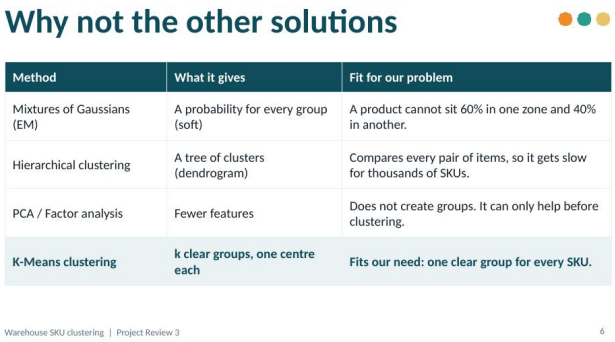
If k is 3, K-Means makes 3 groups. It is like dividing a class into 3 teams so that students with similar marks are in the same team.

IF THE REVIEWER ASKS

Q: What is the weakness of K-Means?

A: We must choose k before starting, and the features must be numbers on a similar scale. We handle both: the elbow method for k, and scaling for the features.

Why not the other solutions



TERMS TO KNOW

Probability: A number from 0 to 1 that says how likely something is.

Pair-wise comparison: Checking every item against every other item.

Feature reduction: Replacing many features with a few.

Zone: One storage area of the warehouse.

SAY THIS

Thank you, Roopesh. You may ask why we did not choose the other methods.

Mixtures of Gaussians give a probability for every group. But a product cannot be kept 60% in one zone and 40% in another. We need one clear zone.

Hierarchical clustering compares every pair of items, so it becomes slow for thousands of SKUs. **PCA and factor analysis** do not create groups at all. They can only help before clustering.

K-Means gives k clear groups with one centre each. That is exactly what we need. Next, how K-Means works.

SIMPLE EXAMPLE

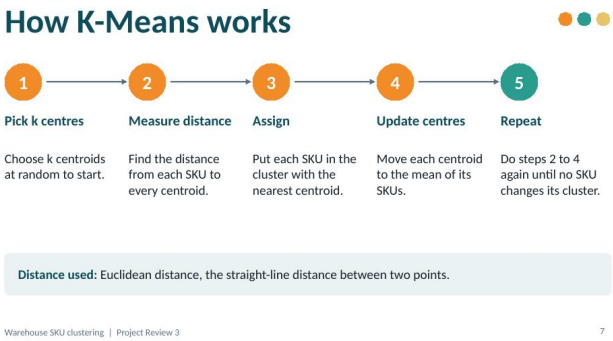
With 1,000 SKUs there are about 5 lakh pairs to compare (1,000 x 999 / 2 = 4,99,500). K-Means only compares each SKU with the k centres, for example 1,000 x 4 = 4,000 distances in one round.

IF THE REVIEWER ASKS

Q: Can PCA still be used?

A: Yes, as a first step. If we had very many features, PCA could reduce them, and then K-Means would make the groups.

How K-Means works



TERMS TO KNOW

Euclidean distance: Straight-line distance. For (1, 2) and (4, 6): square root of (3^2 + 4^2) = 5.

Mean: The average of the values.

Iteration: One round of the repeated steps.

Convergence: The point where nothing changes any more, so we stop.

SAY THIS

K-Means works in five steps.

Step one: choose k centroids at random. Step two: measure the distance from each SKU to every centroid. Step three: assign each SKU to the nearest centroid.

Step four: move each centroid to the mean, that is the average, of the SKUs in its cluster. Step five: repeat steps two to four until no SKU changes its cluster.

The distance we use is **Euclidean distance**, which is the straight-line distance between two points.

SIMPLE EXAMPLE

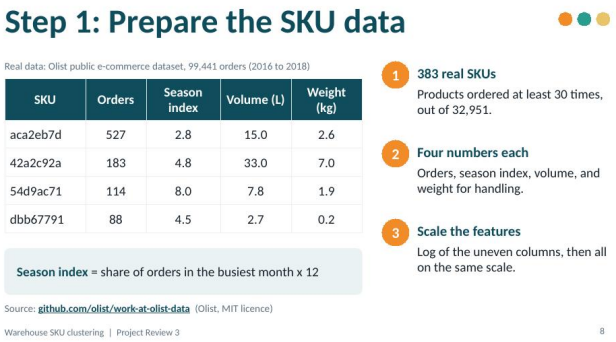
Three teachers stand in a playground. Every student walks to the nearest teacher. Then each teacher moves to the middle of their group. Students again walk to the nearest teacher. When nobody moves any more, the groups are final.

IF THE REVIEWER ASKS

Q: When does the algorithm stop?

A: When no SKU moves to another cluster, or after a fixed number of rounds.

Step 1: Prepare the SKU data



TERMS TO KNOW

Dataset: A table of real records. Ours is Olist's order data.

Season index: Share of orders in the busiest month x 12.

Log scale: Squeezes very large values so a few giants do not dominate.

Scaling: Bringing all features to the same range.

SAY THIS

Now we apply K-Means to real data. We used the public e-commerce dataset from Olist, a Brazilian online marketplace. It has 99,441 real orders from 2016 to 2018. The source link is on the slide.

We kept the 383 products that were ordered at least 30 times. Each SKU becomes one row of four numbers: orders, season index, volume and weight.

Season index is the share of orders in the busiest month, times 12. If it is 1, the product sells the same every month. If it is 8, most orders came in one month. The dataset has no handling field, so **weight** stands for handling needs.

Finally we **scale** the features. Orders, volume and weight are very uneven, so we first take their log, and then bring every column to the same scale.

SIMPLE EXAMPLE

Scaling is like converting marks out of 50 and marks out of 100 into percentages before comparing two students. After that, both subjects count equally.

IF THE REVIEWER ASKS

Q: Where did the data come from?

A: From Olist's public dataset on GitHub: github.com/olist/work-at-olist-data. Our Colab notebook reads it live from that link.

The real data at a glance



TERMS TO KNOW

Order line: One product inside one order.

Black Friday: A big sale day in November, when orders jump.

Filter: Keeping only the rows that meet a rule: here, 30 orders or more.

SAY THIS

Before clustering, let us look at the real data. The dataset has 99,441 orders for 32,951 products, placed from 2016 to 2018.

The graph shows the orders in each month. Orders grow through 2017. The tallest bar is November 2017, the Black Friday month.

From all these products we kept the 383 that were ordered at least 30 times. With fewer orders, we cannot say whether a product is seasonal or not.

SIMPLE EXAMPLE

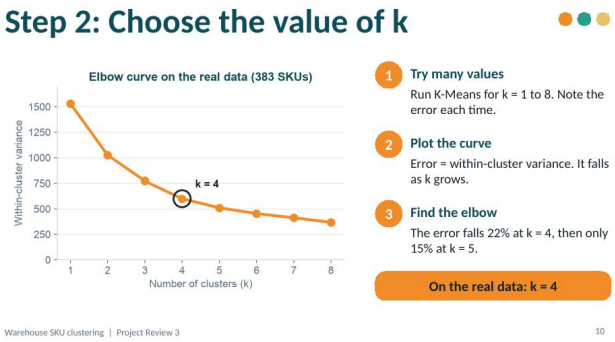
It is like judging a batsman. After 2 matches you cannot say much. After 30 matches you can see his pattern.

IF THE REVIEWER ASKS

Q: Why only 383 out of 32,951 products?

A: Most products were ordered only a few times. With so few orders the seasonal number is not reliable, so we kept the products with 30 or more.

Step 2: Choose the value of k



TERMS TO KNOW

Elbow method: Pick the k where the error curve bends.

Within-cluster variance: How spread out the items are inside their own cluster.

WCSS: Within-Cluster Sum of Squares: the usual name for this error.

Optimal k: The best number of clusters for the data.

SAY THIS

K-Means needs the value of k before it starts. We use the **elbow method** from Unit 3, on the real data.

We ran K-Means for k equal to 1 up to 8 and noted the error each time. The error is the within-cluster variance: how far the SKUs are from their own centroid.

The error falls by 33 percent, then 25, then 22 percent as we go to k equal to 4. After that it falls only 15 percent. So the curve bends at 4.

So the elbow says 4 clusters. Now Rupesh will show a second check.

SIMPLE EXAMPLE

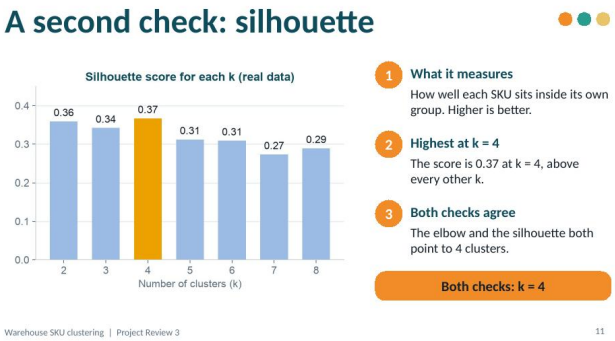
A shop adds billing counters. Going from 1 counter to 4 makes the queue much shorter. Going from 4 to 8 helps only a little. The useful stopping point is 4. That is the elbow.

IF THE REVIEWER ASKS

Q: Is this curve from real data?

A: Yes. It is the error of K-Means on the 383 real SKUs. Anyone can run it again in our Colab notebook.

A second check: silhouette



TERMS TO KNOW

Silhouette score: How well items fit their own cluster. Higher is better; the maximum is 1.

Cohesion: How close the items inside one cluster are.

Separation: How far one cluster is from the others.

SAY THIS

Thank you, Charith. The elbow gave us k equal to 4. We checked it in a second way, with the **silhouette score**.

The silhouette score tells how well each SKU sits inside its own group, compared with the nearest other group. Higher is better, and the most it can be is 1.

In the graph, the tallest bar is at k equal to 4, with a score of 0.37. So the elbow and the silhouette agree: 4 clusters.

SIMPLE EXAMPLE

Think of students at lunch. If everyone sits clearly with their own friend group, the score is high. If many sit between two groups, the score is low.

IF THE REVIEWER ASKS

Q: Is 0.37 a good score?

A: It is moderate, which is normal for real data. What matters is that k = 4 gives the highest score of all the values we tried.

Step 3: Clusters become zones



TERMS TO KNOW

Fast mover: A product that is picked very often.

Slotting: Deciding the storage place of each product.

Zone: One storage area of the warehouse.

Dispatch: Sending the packed order out.

SAY THIS

K-Means found four clusters in the real data.

Cluster A is **fast movers**: 76 SKUs, ordered about 160 times each. They go on the front racks, nearest to packing. Cluster B is **bulky and heavy**: 84 SKUs, about 47 litres and 7.6 kilos on average. They stay at ground level.

Cluster C is **seasonal**: 25 SKUs whose orders bunch into one month. They go to a flexible zone and move forward in season. Cluster D is **regular small items**: 198 SKUs, on the back racks.

The floor plan on the right shows this. Fast movers are closest to packing and dispatch.

SIMPLE EXAMPLE

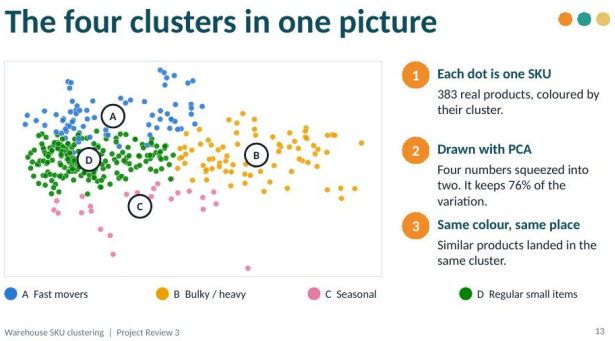
It is like a kitchen. Salt and oil are next to the stove. The big rice bag stays on the floor. Festival vessels are kept at the back, and brought forward at festival time.

IF THE REVIEWER ASKS

Q: Who named the clusters?

A: We did, from the averages of each cluster. K-Means only gives the groups; people read them and name them.

The four clusters in one picture



TERMS TO KNOW

PCA: Principal Component Analysis: reduces many numbers to a few.

Principal component: A new axis that carries the most variation in the data.

Variation kept: How much of the original information survives: here 76%.

SAY THIS

This picture shows all 383 SKUs. Every dot is one product, and its colour is its cluster.

A SKU has four numbers, so we cannot draw it directly. We used **PCA**, which is also in Unit 3. PCA squeezes the four numbers into two, and still keeps 76 percent of the variation.

Dots of the same colour sit together. That means similar products landed in the same cluster. The letters A, B, C and D mark the cluster centres.

SIMPLE EXAMPLE

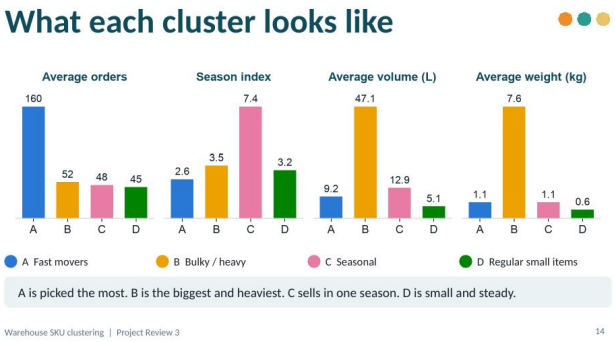
PCA is like taking a photo of a statue. The statue has three dimensions and the photo has two, but from a good angle you still see its shape.

IF THE REVIEWER ASKS

Q: Did PCA make the clusters?

A: No. K-Means made the clusters using all four numbers. PCA was used only to draw them in two dimensions.

What each cluster looks like



TERMS TO KNOW

Average (mean): The sum of the values divided by how many there are.

Cluster profile: The averages that describe one cluster.

Season index: Share of orders in the busiest month x 12.

SAY THIS

These four graphs show the average of each number, cluster by cluster.

Cluster A is ordered the most, about 160 times, so it is the **fast movers**. Cluster B has the biggest volume and weight, about 47 litres and 7.6 kilos, so it is **bulky and heavy**.

Cluster C has the highest season index, 7.4, so it is **seasonal**. Cluster D is small and steady, so we call it **regular small items**.

This is how the clusters got their names.

SIMPLE EXAMPLE

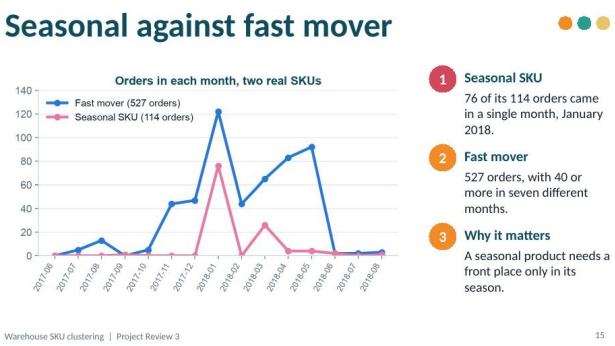
It is like reading a class report. One section has the highest sports score, another the highest maths score. We name each section by what stands out.

IF THE REVIEWER ASKS

Q: Why does cluster B come second?

A: Zones are ordered by average orders. B is ordered a little more than C and D, so it comes second.

Seasonal against fast mover



TERMS TO KNOW

Seasonal SKU: A product whose orders bunch into a short period.

Fast mover: A product that is ordered often.

Peak month: The month with the most orders.

SAY THIS

Here are two real products, month by month.

The pink line is the most ordered SKU of the seasonal cluster. It has 114 orders, and 76 of them came in a single month, January 2018.

The blue line is the most ordered fast mover. It has 527 orders, with 40 or more in seven different months.

So a seasonal product needs a front place only in its season, but a fast mover needs it for many months. Now Anbu will show how this solves our problem.

SIMPLE EXAMPLE

Crackers sell in a rush before Diwali. Milk sells every day. A shop keeps milk in front always, and brings crackers to the front only in the season.

IF THE REVIEWER ASKS

Q: Is one peak month enough to call it seasonal?

A: The data covers only about two years, so we can say its orders are bunched into one month. With more years we could confirm that it repeats every year.

How this solves our problem



TERMS TO KNOW

Travel time: The time a worker spends walking to collect items.

Model: A simplified calculation that stands in for the real thing.

Typical product: What the centroid of a cluster describes.

SAY THIS

Thank you, Rupesh. How does this solve our problem? Our goal was better placement and less travel time.

First, **less walking**. Fast movers sit closest to packing. In our simple one-aisle model, the walk per pick falls from about 64 metres to about 47 metres. That is 27 percent less.

Second, **better placement**. Bulky and heavy products get their own ground-level zone. Third, **season ready**. Seasonal products are one group, so they are easy to move forward in season.

One honest point: the walking figure is a model, not a measurement in a real warehouse.

SIMPLE EXAMPLE

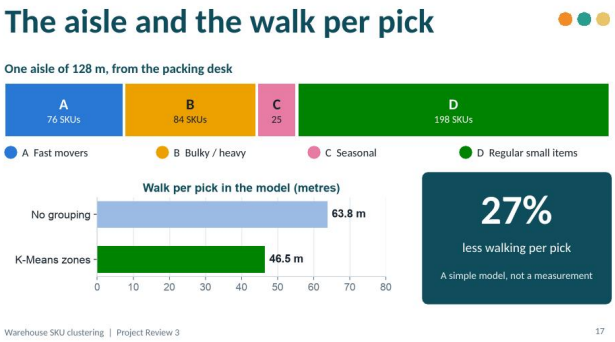
The centroid of the fast movers says: about 160 orders, 9 litres, 1 kilo. We can read it as a typical small, popular product.

IF THE REVIEWER ASKS

Q: Is 27 percent a real saving?

A: It is what our simple model gives on the real order data. A real warehouse would need its own layout and measurements.

The aisle and the walk per pick



TERMS TO KNOW

Aisle: The walking path between racks.

Walk per pick: The average distance a worker walks to collect one item.

Model: A simplified calculation that stands in for the real thing.

SAY THIS

The coloured strip is one aisle of 128 metres, starting at the packing desk. Each cluster takes as much of the aisle as it has products.

Cluster A, the fast movers, comes first, in the first 25 metres. Cluster D, the regular small items, is the largest and takes the far end.

The bars show the walk for each pick. With no grouping it is about 64 metres. With the K-Means zones it is about 47 metres. That is 27 percent less.

This is a simple model, not a measurement in a real warehouse.

SIMPLE EXAMPLE

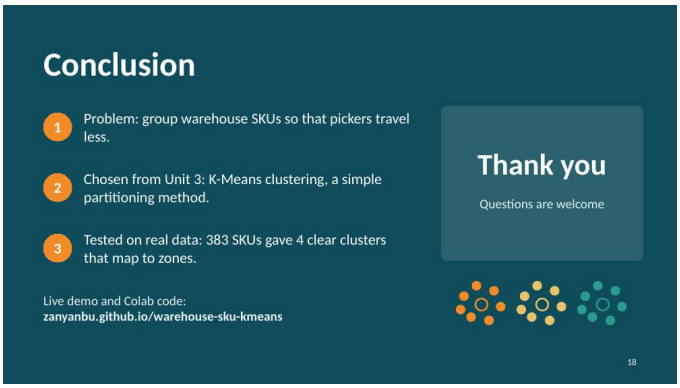
In a cupboard, the things used daily are kept near the door, and the things used rarely are kept deep inside. You reach less, because what you need most is nearest.

IF THE REVIEWER ASKS

Q: How did you get 64 metres for no grouping?

A: With no grouping a product can be anywhere, so on average it is half-way down the 128 metre aisle. Half of 128 is 64.

Conclusion



TERMS TO KNOW

Partitioning method: A method that divides data into separate groups.

Outcome: What the solution gives us in the end.

SAY THIS

To conclude. Our problem was to group warehouse SKUs so that pickers travel less.

From Unit 3 we selected **K-Means clustering**, because it is a simple partitioning method that gives every SKU one clear group.

We tested it on real data. 383 SKUs gave 4 clear clusters that map directly to warehouse zones.

The live demo and the Colab code are at the link on the slide. Thank you. We are happy to take your questions.

SIMPLE EXAMPLE

In one line: similar products together, fast products nearest, and the grouping done by data instead of by guesswork.

IF THE REVIEWER ASKS

Q: If you had more time, what would you add?

A: A warehouse's own data with real handling labels, and a measured walking distance before and after.

All the terms in one place

Every speaker should know these, whoever's slide the question comes on.

Centroid	The centre (mean point) of a cluster.	**k**	The number of clusters we ask for.
Cluster	A group of similar data points.	**Mixture of Gaussians**	Soft clustering with probabilities.
Clustering	Dividing data into groups of similar items.	**PCA**	Reduces many features to a few.
Convergence	When nothing changes any more and the algorithm stops.	**Pick frequency**	How often a product is picked.
Dendrogram	A tree diagram from hierarchical clustering.	**Scaling**	Bringing features to the same range.
Elbow method	Choosing k where the error curve bends.	**Season index**	Share of orders in the busiest month x 12.
EM algorithm	Expectation-Maximization: guess, improve, repeat.	**SKU**	Stock Keeping Unit: one product type.
Euclidean distance	Straight-line distance between two points.	**Soft clustering**	Each item gets a probability per group.
Feature	A measurable property of an item.	**Unsupervised learning**	Learning from data without labels.
Hard clustering	Each item is in exactly one group.	**Within-cluster variance**	How spread out items are inside a cluster.

Questions any speaker may get

Q: Why K-Means and not mixtures of Gaussians?
A: A product needs one fixed zone. K-Means gives one group per SKU; a mixture gives probabilities.

Q: How did you choose k?
A: With the elbow method: plot the error for many k values and pick the bend.

Q: Which distance do you use?
A: Euclidean distance, the straight-line distance, after scaling the features.

Q: What are the limits of K-Means?
A: k must be chosen first, features must be scaled numbers, and the result depends on the random start, so we run it more than once.

Q: Are your numbers real?
A: Yes. Orders, sizes and weights come from Olist's public dataset (github.com/olist/work-at-olist-data). Only the walking distance is a simple model.